

Technology, Automation, AI, and Labor Market

Labor II — Week 11

Teaching deck draft

Central question and course repositioning

- How do technology, automation, and AI reshape work through worker adaptation, firm labor demand, and market-level reallocation?
- Week 11 is a major shock-and-adjustment lecture, not a generic AI detour.
- The labor-market objects stay familiar: tasks, employment, wages, vacancies, worker learning, and equilibrium incidence.

Week 10 to Week 11 bridge

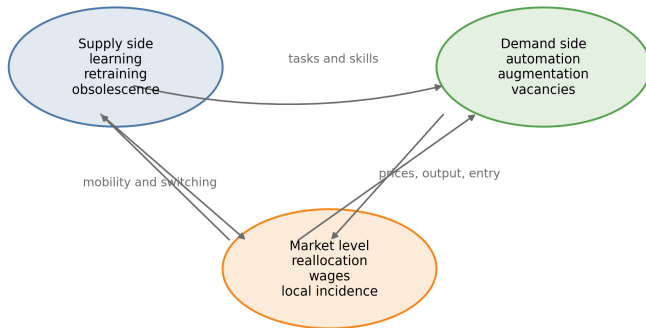
- Week 10: cyclical shocks moved unemployment, vacancies, separations, mobility, and match quality.
- Week 11: technology shocks move many of the same margins, but through more persistent restructuring.
- Week 12 will then shift from technology shocks inside production to trade and offshoring shocks through global integration.

Why technology is not just a labor-demand topic

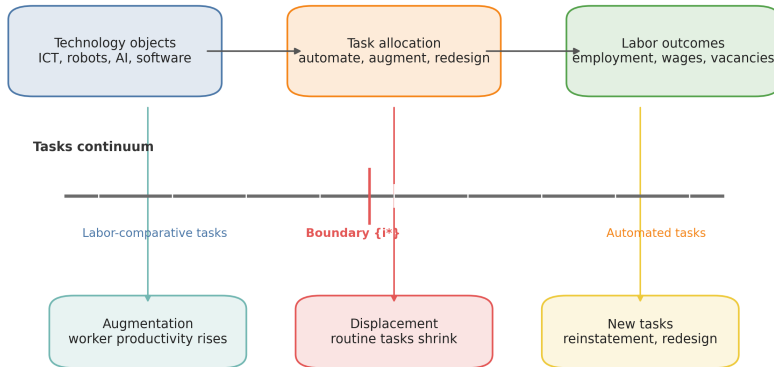
- Supply side: learning, retraining, search/application technology, and skill obsolescence.
- Demand side: automation, augmentation, vacancy redesign, and workforce composition.
- Market level: local reallocation, labor-share shifts, wage incidence, and cross-place spillovers.

Supply, demand, and market-level feedbacks

Technology shocks propagate through linked labor-market margins



Task-based framework: automation, augmentation, and new tasks



One task-based production object

$$Y_t = \left[\int_0^1 y_t(i)^{\frac{\sigma-1}{\sigma}} di \right]^{\frac{\sigma}{\sigma-1}}$$

- Tasks, not occupations, are the primitive object.
- Technology changes task productivity and the elasticity of reallocation across tasks.
- Observed occupations are bundles of tasks with changing internal content.

Automation boundary and task allocation

$$\frac{w_t}{a_t(i_t^*)} = \frac{c_t^T}{b_t(i_t^*)}, \quad \mathcal{A}_t = \left\{ i : \frac{c_t^T}{b_t(i)} \leq \frac{w_t}{a_t(i)} \right\}$$

- Lower technology cost or higher machine productivity expands the automated task set.
- That shift does not by itself reveal the sign of aggregate labor-demand effects.
- Productivity, output expansion, and new tasks can offset direct displacement.

Displacement, productivity, and reinstatement

$$\Delta \log L_t = \Delta \log L_t^{disp} + \Delta \log L_t^{prod} + \Delta \log L_t^{new}$$

- Direct substitution is only one term in the labor-demand accounting.
- Empirically, students should ask which term a paper is closest to observing.
- Market-level incidence depends on all three channels plus reallocation frictions.

Supply-side adjustment: learning, training, and obsolescence

$$H_{t+1} = (1 - \delta_t)H_t + I_t$$

- Technology can raise the payoff to learning while increasing skill obsolescence.
- Worker-side adjustment is not just schooling; it includes on-the-job learning and occupational switching.
- Productivity gains for some workers need not imply welfare gains for all workers.

Historical perspective: ICT, polarization, and long-run adjustment

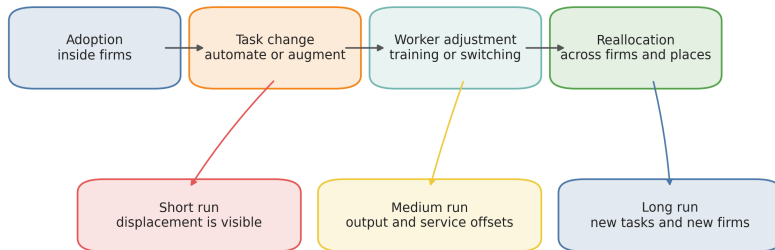
- Autor: labor demand did not disappear because technology also created complementary and new tasks.
- Acemoglu-Restrepo: automation displaces labor, but reinstatement matters.
- Michaels-Natraj-Van Reenen: ICT shifted demand toward high-skill tasks and away from routine middle-skill work.

Demand-side evidence: robots, ICT, AI adoption, and vacancies

- Robots: local-labor-market exposure captures wage and employment incidence across places.
- Cross-country robot panels capture broad productivity and labor-share patterns.
- Vacancy data reveal changing task content and hiring intent, not realized employment by themselves.
- Firm adoption data are closer to treatment, but selection into adoption is central.

Market-level equilibrium and reallocation

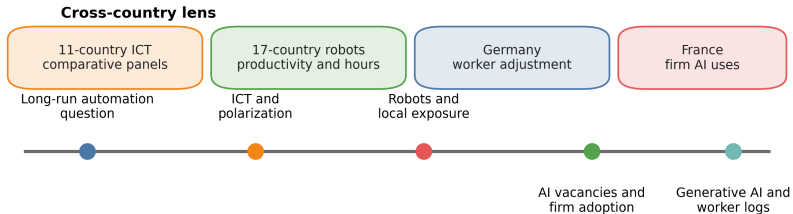
Adjustment depends on time horizon and mobility frictions



Contemporary AI evidence

- Felten-Raj-Seamans: occupation-level AI exposure is a descriptive susceptibility measure.
- Acemoglu-Autor-Hazell-Restrepo: online vacancies show changing demand and task content.
- Aghion et al.: different uses of AI inside firms can produce different labor-demand responses.
- Brynjolfsson-Li-Raymond: worker productivity effects are not the same object as market-level employment effects.

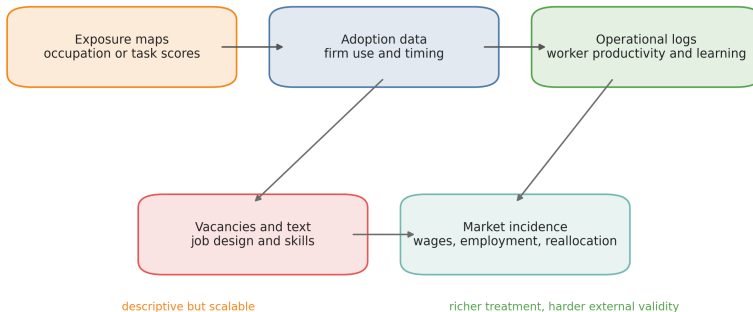
Historical and global evidence map



Compare eras by observed margin, identifying design, and institutional setting

AI data and labor-market measurement

As treatment measurement improves, the equilibrium object often narrows



Reduced-form exposure equation and design discipline

$$\Delta Y_{mt} = \beta TechExposure_{mt} + X'_{mt}\Gamma + \varepsilon_{mt}$$

- Ask whether the technology object is exposure or realized adoption.
- Ask whether the unit is a worker, firm, occupation, industry, or local market.
- Ask which labor margin is observed and which equilibrium margin stays offstage.

Historical evidence versus very recent AI evidence

- ICT and robots come with longer diffusion horizons and clearer realized capital adoption objects.
- Very recent AI evidence has stronger exposure measures and faster-moving firm narratives than long-run outcome data.
- Students should avoid narrating descriptive exposure maps as if they were already causal adoption evidence.

Frontier research questions

- Exposure versus adoption measurement
- AI and worker-side search/application technology
- Organizational complements and the productivity J-curve
- Technology, wage-setting, rent sharing, and labor share
- Gender, inequality, and place-based incidence
- Training, diffusion policy, and worker transition support

Bridge to Week 12 trade and offshoring

- Week 11: shocks begin inside production, task choice, and organizational redesign.
- Week 12: shocks begin through global integration, import competition, and offshoring.
- Both weeks are about labor-market adjustment, but the mapping from exposure to incidence differs.

- Technology is not just a labor-demand topic; it is a supply, demand, and equilibrium topic.
- Tasks are more primitive than occupations for theory and measurement.
- Exposure, adoption, within-firm evidence, and market-level incidence are distinct objects.
- The core empirical challenge is to separate displacement from productivity, new-task, and reallocation effects.

Appendix: Week 11 lab bridge

- Reproduce: a bounded firm-level AI-use design inspired by Aghion et al.
- Diagnose: adoption versus exposure and which margin is actually observed.
- Transfer: move one exposure idea to a small synthetic market-level setting.
- Challenge extension: contrast within-firm AI evidence with a robot-style local-labor-market design.